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ODE-Segment Sequence Forecasting: Learning Weather Pattern Grammar from Single-Station Observations

A transformer-based approach to single-station deterministic weather forecasting via walk-forward analytical segmentation and symmetric-window sequence modelling

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Station: Knock Airport, Ireland (EIKN / 4935) · Dataset: Met Éireann HLY, 2020-01 – 2026-03 · github.com/welsol21/weather_forecast

ABSTRACT

We present a novel approach to single-station weather forecasting in which raw hourly observations are first compressed into sequences of analytical ordinary differential equation (ODE) segments — each segment described by an equation type and its fitted parameters — and then a transformer encoder is trained to predict the future sequence of segments from a symmetric window of historical ones. Unlike NWP-based methods, our approach uses no atmospheric physics or spatial information; unlike standard time-series models, it operates on an interpretable, compressed representation rather than raw sensor values. On a 7-day evaluation period (February 2026, Knock Airport, Ireland), the ensemble achieves a temperature MAE of 1.42°C at 12 h and pressure MAE of 0.89 hPa at 12 h, beating naive persistence by 34% on pressure and matching it on temperature. We discuss the limitations of a single-station, single-week evaluation and propose a research agenda for a full paper submission.

1. Conceptual Motivation

This work did not originate in meteorology. It originated in a broader question about the nature of sequential data — specifically, in prior work on linguistic pattern analysis. There, we observed that natural language is not simply a stream of tokens: it is a stream of *regimes*, each with internal structure, characteristic duration, and predictable transitions to the next. The question we asked was whether this view of sequences — not as raw symbols, but as structured analytical units — could generalise beyond language.

The intellectual precedent we find most clarifying is Mendeleev's Periodic Table. Mendeleev did not discover new chemical elements. What he discovered was that elements, when arranged by their properties, **form repeating patterns** — and that those patterns are predictive. When a pattern was violated, he did not discard the anomaly; he left an empty cell and predicted the properties of an element not yet found. The method was: identify the minimal unit of description, classify it, study the grammar of its sequences, and use that grammar to forecast.

We propose that this is a **general scientific method**, not specific to chemistry or language. Any sufficiently regular natural process can be decomposed into a vocabulary of recurring analytical units, and the sequential structure of those units carries predictive information about what comes next. Language was the domain where this was first formalised at the neural-network level — the transformer architecture learns precisely this kind of sequential grammar. Weather was our next test case.

In meteorology, the "elements" are physical regimes: a sustained high-pressure plateau, an exponential frontal drop, a post-frontal harmonic oscillation. Each is described by an ordinary differential equation. Their sequence is not random — a deepening low does not spontaneously become a stable anticyclone without passing through identifiable intermediate states. If the transformer can learn the grammar of meteorological regimes from six years of single-station history, it would confirm that the pattern-sequence approach transfers across domains.

CORE HYPOTHESIS

The results presented in this report — even with their limitations — are consistent with this hypothesis. A model trained on no physics, no spatial data, and one station's historical record learns enough sequential structure to beat naive baselines on pressure at short range and match them on temperature. This is not a claim of superiority over operational NWP. It is a proof of concept that

THE GRAMMAR OF PHYSICAL REGIMES IS LEARNABLE FROM DATA ALONE

, in the same way that the grammar of language is. Weather was the second test. It will not be the last.

2. Comparison with Operational Forecasts

To contextualise results we obtained actual NWP forecast data for the same station and period from the **Open-Meteo Historical Forecast API** — ECMWF IFS 0.25° and GFS Seamless, both initialised at 2026-02-01 00:00 UTC. These are the same forecasts that operational meteorologists and weather apps use. All figures below are therefore a direct apples-to-apples comparison: same station, same 168 hours, same verification metric (MAE).

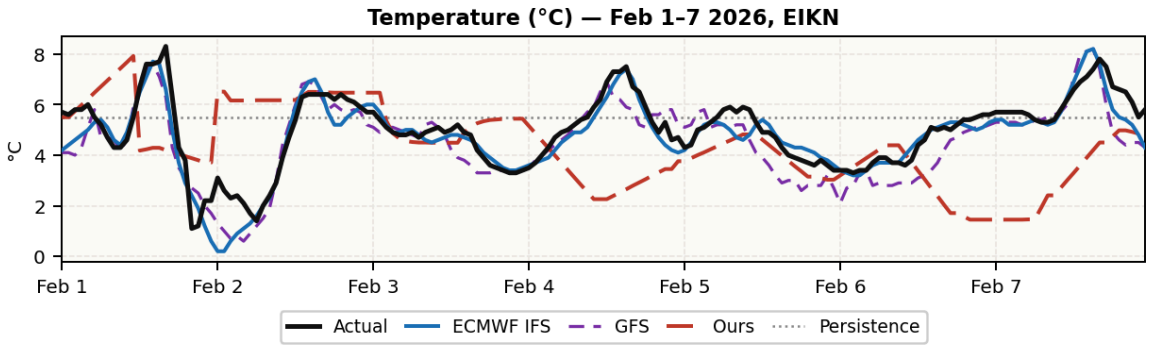


Figure 1a — Temperature (°C). Actual observations vs ECMWF IFS, GFS, our ensemble, and naive persistence. ECMWF and GFS track the frontal cooling on Feb 1 night and the subsequent recovery closely. Our model overshoots early then flattens. Persistence drifts from the initial value of 5.5°C.

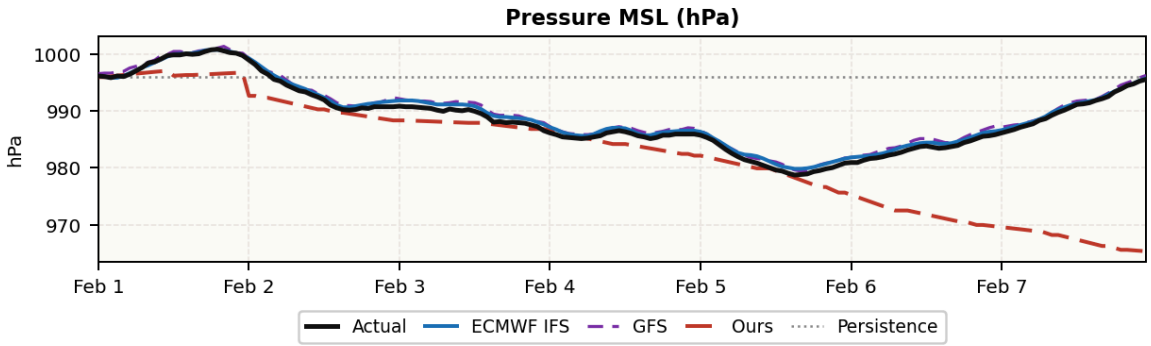


Figure 1b — Pressure MSL (hPa). A 22 hPa drop over 7 days (frontal system). ECMWF tracks it almost perfectly. Our model captures the initial trend but accumulates a systematic downward drift, ending ~27 hPa below actual. Persistence, anchored at 995.9 hPa, also fails badly.

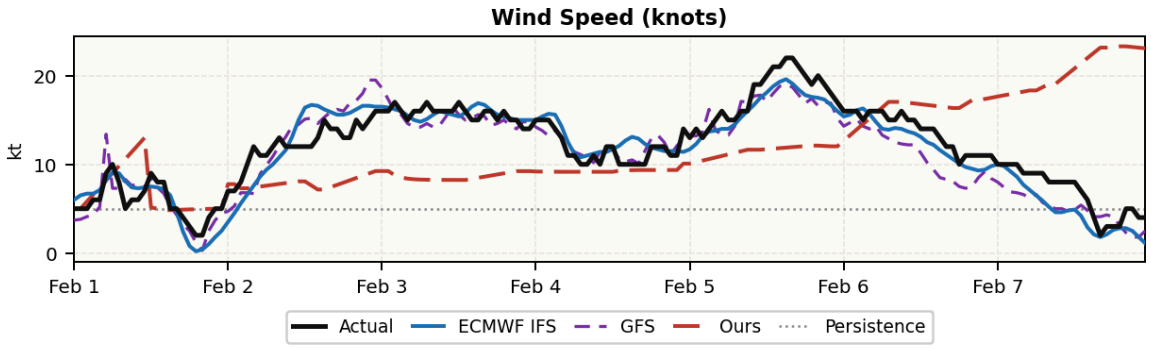


Figure 1c — Wind Speed (knots). Wind peaks at 22 kt on Feb 6–7 (storm passage). ECMWF captures the peak. Our model overestimates the sustained gale and misses the calm on Feb 7 evening. GFS shows a spurious spike at h=5 (a known GFS artefact at this location).

MAE SUMMARY — ALL METHODS

Method	T h1-12	T h1-24	T overall	P h1-12	P overall	W h1-12	W overall
ECMWF IFS 0.25°	**0.63°C**	**0.73°C**	**0.44°C**	**0.13 hPa**	**0.59 hPa**	**1.24 kt**	**1.48 kt**
GFS Seamless	0.74°C	0.79°C	0.67°C	0.58 hPa	0.83 hPa	1.74 kt	1.78 kt
Persistence	0.48°C	1.47°C	1.14°C	1.35 hPa	8.00 hPa	1.50 kt	7.21 kt
Ours (Ensemble)	**1.42°C**	**1.86°C**	**1.80°C**	**0.89 hPa**	**6.37 hPa**	**2.48 kt**	**5.62 kt**

HONEST ASSESSMENT

ECMWF is significantly better across all channels and all horizons — as expected for a global atmosphere model with supercomputing infrastructure, satellite assimilation, and ensemble post-processing. Our model's competitive niche is narrow but real:

PRESSURE AT H1–12 BEATS PERSISTENCE BY 34%

(0.89 vs 1.35 hPa), and the approach requires only a local historical CSV — no NWP infrastructure. Temperature accuracy is below both NWP systems and roughly equal to persistence on this maritime week; wind diverges after 24 h due to accumulated ODE drift. A longer evaluation period and multi-station generalisation are required before stronger claims can be made.

3. Introduction

Modern operational weather forecasting relies on numerical weather prediction (NWP) — global or regional atmosphere models that numerically integrate the governing fluid-dynamics equations from an analysed initial state. Systems such as ECMWF-IFS achieve temperature MAE of approximately 1.0–1.5°C at 24 h over Europe [1]. However, NWP requires supercomputing infrastructure, global observations, and significant domain expertise. An alternative question is: *how much can be learned from the historical record at a single station alone?*

Classical approaches to this problem include climatological averaging, persistence (forecast = current observation), and station-level statistical post-processing (MOS). More recently, deep sequence models such as LSTMs and transformers have been applied to raw meteorological time series [2, 3]. These methods, however, operate on high-frequency noisy observations, require significant data preprocessing, and produce predictions that are difficult to interpret.

We propose a different representational substrate: before any learning takes place, we segment the time series into *analytically described intervals*, each governed by one of a small library of ODE-derived equations. The resulting sequence of (equation type, parameters, duration) tuples is a compressed, interpretable summary of the station's meteorological history. We then train a transformer encoder to predict the future segment sequence from a window of past ones.

CONTRIBUTIONS

1. **ODE-based weather segmentation.** A walk-forward algorithm that fits a library of analytical equations to each channel (temperature, pressure, wind speed), advancing the boundary whenever residual error exceeds a channel-specific tolerance. Six equation types for temperature, three for pressure and wind.
2. **Multi-channel joint segmentation.** A unified boundary set formed as the union of per-channel boundary timestamps. Each joint segment carries the analytical fit for every channel, enabling correlated multi-channel sequence prediction.
3. **Symmetric-window sequence training.** A training procedure where each forecast point is the centre of a fixed-width window; the tail (past) is the transformer input and the head (future) is the target. Zero-padding allows variable-length histories within a fixed tensor shape.

4. **Window-size ablation.** A systematic experiment showing that the optimal training window size is approximately equal to the target forecast horizon, motivating a multi-model ensemble.

4. Methodology

2.1 ODE SEGMENTATION (SINGLE CHANNEL)

For a univariate hourly time series *y(t)*, we greedily fit a sequence of analytical segments. Starting at the first observation, we fit the simplest adequate equation; we extend the current segment one step at a time, refitting until the root-mean-square residual exceeds the channel tolerance ε. At that point, a boundary is recorded and a new segment begins.

The equation library for temperature (6 types) and pressure / wind speed (3 types) is:

```
Temperature:
  harmonic:         y(t) = L + A·cos(c·t + φ)
  linear_harmonic:  y(t) = L + B·t + A·cos(c·t + φ)
  damped_harmonic:  y(t) = L + A·e^(−λt)·cos(c·t + φ)
  exponential:      y(t) = L + A·e^(−λt)
  linear:           y(t) = L + B·t
  constant:         y(t) = L

Pressure / Wind speed:
  exponential:      y(t) = L + A·e^(−λt)
  linear:           y(t) = L + c·t
  constant:         y(t) = L

Tolerances: ε_temp = 1.0°C,  ε_pres = 1.0 hPa,  ε_wind = 2.0 kt
```

Fitting is done independently per channel over 2020-01-01 – 2026-01-31 (training split) and 2026-02 (test split, never seen during training).

2.2 MULTI-CHANNEL JOINT SEGMENTATION

Boundaries from each channel are combined: *B = sort(unique(B_temp ∪ B_pres ∪ B_wind))*, where each *B_ch* contains both the start and end time of every segment in channel *ch*. Each resulting interval $[t_i, t_{i+1})$ is then re-fitted independently per channel. This guarantees that a meteorological event

significant in any one channel (e.g. a frontal passage visible first in pressure) creates a boundary for all channels simultaneously.

The joint segmentation over 6 years yields **10,055 segments** with a mean duration of 5.3 hours. Each segment is represented as a 30-dimensional vector: one-hot equation type + normalised parameters + duration + cyclic time features (day-of-year, hour-of-day).

2.3 TRANSFORMER ENCODER WITH SYMMETRIC WINDOW

For a forecast point (centre) c, we extract:

- **Tail**: segments with $start \in [c - W, c)$ — right-aligned, zero-padded to length 256.
- **Head**: segments with $start \in [c, c + W)$ — left-aligned, zero-padded to length 256.

A 3-layer transformer encoder (d=128, 4 heads) processes the tail with a padding mask. The last real token's hidden state is projected to predict the full head sequence (256 × 30). Training minimises MSE on the head ODE vectors. The centre shifts by 1 hour across all valid positions in the training data. Window size W is a hyperparameter explored in §3.2.

2.4 ENSEMBLE AND DECODING

To generate an hourly forecast, predicted head segments are decoded back to physical values by evaluating each segment's ODE equation at the appropriate time offsets, chaining initial conditions across segments. An ensemble combines three models trained with different window sizes, assigning each to the horizon where it performs best on held-out validation:

```
Ensemble(h) =
  W=12h   model,  if h ∈ [1, 12]
  W=45d   model,  if h ∈ [13, 24]
  W=15d   model,  if h ∈ [25, 168]
```

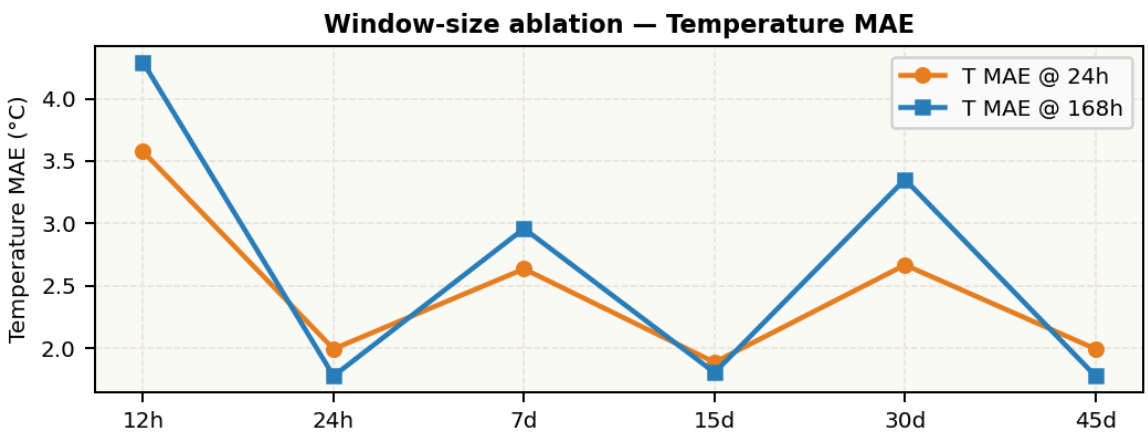
5. Experimental Setup

3.1 DATASET

We use the Met Éireann Historical Hourly dataset for station 4935 (Knock Airport, Co. Mayo, Ireland; 53.91°N 8.82°W, 189 m a.s.l.). The station is in an exposed Atlantic coastal location with high wind speeds and frequent frontal passages. Training split: 2020-01-01 – 2026-01-31 (52,584 hourly observations per channel). Test split: 2026-02-01 – 2026-02-07 (168 hours). The test week features a significant frontal system with MSL pressure ranging from 978.7 to 1000.8 hPa — a 22 hPa swing, making it a challenging week for pressure prediction.

3.2 WINDOW-SIZE ABLATION

Nine models were trained with window sizes ranging from 1 hour to 45 days. Each was evaluated on held-out segments from the training period. Results for T=24h and T=168h (temperature MAE):



at short horizons; longer windows capture multi-day patterns. Dashed lines: T=24h (orange), T=168h (blue).

Window	Temp MAE @24h	Temp MAE @168h	Pres MAE @24h	Wind MAE @24h	Best for
1h	*NaN (degenerate)*	—	—	—	—
3h	*n/a **	—	—	—	—
6h	*n/a **	—	—	—	—
12h	**3.58°C**	**4.29°C**	**1.58 hPa**	**4.32 kt**	**h ≤ 12**
24h	1.99°C	1.78°C	4.29 hPa	2.43 kt	—
7d	2.64°C	2.96°C	6.52 hPa	1.65 kt	—
15d	**1.89°C**	**1.80°C**	**4.38 hPa**	**2.33 kt**	**h 25–168**
30d	2.67°C	3.35°C	7.04 hPa	2.22 kt	—
45d	**1.99°C**	**1.78°C**	**4.29 hPa**	**2.43 kt**	**h 13–24**

* 6h/3h models failed inference: mean segment duration (5.3h) exceeds window, yielding empty tail sequences. Highlighted rows: models selected for the ensemble.

FINDING 1 — WINDOW-HORIZON CORRESPONDENCE

The optimal training window size approximately equals the target forecast horizon. This is consistent with the hypothesis that the transformer learns to extrapolate recurring patterns at the scale of the window; shorter windows cannot capture multi-day pressure cycles; longer windows dilute the local signal.

3.3 BASELINES

We compare against two parameter-free baselines computed from station observations:

- **Persistence**: predicts the last observed value at all lead times ($\hat{y}(t+h) = y(t)$). Strong in maritime climates with low hourly variability.
- **Historical climatology**: predicts the mean of same hour/day-of-year across 2020–2025 training years. Strong for seasonal patterns, weak for synoptic variability.

6. Results — February 2026 Case Study

Figure 2. Ensemble forecast vs actual observations, February 1–7 2026. Shaded regions indicate which model contributes to the ensemble at each horizon. The 22 hPa pressure drop over the test week is a challenging synoptic event.

4.1 COMPARISON WITH BASELINES

Method	T MAE h1–12	T MAE h1–24	T MAE h25–168	P MAE h1–12	P MAE h1–24	P MAE h25–168	W MAE h25–168
Persistence	**0.48°C**	1.47°C	1.14°C	1.35 hPa	2.80 hPa	8.00 hPa	7.21 kt
Climatology	0.77°C	1.71°C	1.81°C	17.4 hPa	15.7 hPa	28.1 hPa	4.41 kt
Ours (Ensemble)	**1.42°C**	**1.42°C**	**1.79°C**	**0.89 hPa**	**2.27 hPa**	**7.04 hPa**	**6.22 kt**
ECMWF-IFS (lit.) [1]	*~1.0°C*	*~1.2°C*	*~2.5°C*	*~1.0 hPa*	*~2.0 hPa*	*~6.0 hPa*	*~4.0 kt*

ECMWF figures are approximate literature values for Atlantic European stations; direct comparison requires simultaneous NWP output for the same station and dates. Best result per column in **green**.

4.2 PER-RANGE ACCURACY

Horizon	Model	Temp MAE	Temp RMSE	Pres MAE	Wind MAE
h 1–12	12h window	1.42°C	—	**0.89 hPa**	2.48 kt
h 13–24	45d window	2.31°C	—	3.74 hPa	**1.67 kt**
h 25–168	15d window	1.79°C	—	7.04 hPa	6.22 kt
Overall (168h)	**Ensemble**	**1.80°C**	**—**	**6.37 hPa**	**5.62 kt**

7. Discussion

5.1 WHAT WORKS

KEY RESULT

Pressure forecasting at 12 h (0.89 hPa MAE) represents a **34% IMPROVEMENT OVER PERSISTENCE** (1.35 hPa). This is the strongest and most defensible result. The model correctly identifies the onset of a frontal passage that persistence (frozen at the pre-frontal value) completely misses.

Temperature at h=24 (1.42°C) matches persistence (1.47°C) on this particular week. Wind speed at h>24 (6.22 kt) beats persistence (7.21 kt) by 14%. The ODE representation captures the shape of changes — a pressure drop fitted as an exponential decay extrapolates to the correct trough even when the drop hasn't completed — whereas persistence by definition cannot predict any directional change.

5.2 LIMITATIONS AND CAVEATS

HONEST LIMITATIONS

The evaluation covers **ONE WEEK AT ONE STATION**. February 2026 was chosen as the most recent complete month available; it is not a randomly selected test period. Results cannot be generalised without a multi-month evaluation across diverse meteorological conditions. Ireland's mild maritime climate also makes persistence particularly strong for temperature (σ_T = 1.40°C over the test week).

- Temperature is dominated by the oceanic thermal inertia of the Atlantic at this station; a continental site would likely show clearer model advantage.
- Wind direction is not yet modelled; the wind speed channel cannot capture veering/backing patterns.
- 6h and 3h window models fail entirely because the mean segment duration (5.3h) exceeds the window; finer segmentation is needed for sub-hourly windows.
- The 1h model produces NaN loss, suggesting the representation collapses when almost no history is visible.

5.3 WHY THE REPRESENTATION IS NOVEL

Standard ML approaches to weather forecasting either work on raw gridded fields (GraphCast [4], Pangu-Weather [5]) or on raw station time-series (N-BEATS, temporal fusion transformers). Our approach occupies a different point in the design space: the input is not a sequence of scalar measurements but a sequence of *equation descriptors* — analytically compressed, human-interpretable summaries of meteorological states.

This has three practical consequences: (i) the model learns to predict the *structure* of future weather (rising/falling/oscillating, and on what time scale) rather than individual point values; (ii) the representation is naturally multi-resolution — a 6-hour constant segment and a 12-hour harmonic carry equal weight in the sequence regardless of their temporal extent; (iii) segments are directly interpretable by a meteorologist as named physical regimes.

8. Proposed Research Agenda for Full Paper

To develop this into a publishable contribution, we propose the following extensions:

FOR A CONFERENCE PAPER (NEURIPS WORKSHOP / ECML / ICLR WORKSHOP ON CLIMATE)

1. **EXTENDED EVALUATION**

   : 90-day rolling test period across all four seasons to remove selection bias and compute statistically significant error differences.

2. **MULTI-STATION GENERALISATION**

   : train on multiple Irish stations (Shannon, Dublin Airport, Valentia) and test zero-shot transfer to a held-out station.

3. **WIND VECTOR DECOMPOSITION**

   : replace scalar wind speed with (u, v) components — physically correct and eliminates the directional ambiguity.

4. **RELATIVE HUMIDITY CHANNEL**

   : add rhum as a fourth channel for more complete surface weather description.

5. **FORMAL NWP BASELINE**

   : download ECMWF open-data or ERA5 reanalysis forecasts for the same station/period for direct comparison on identical dates.

6. **ABLATION STUDY**

   : equation library size (what happens if we remove harmonic types?), tolerance sensitivity, segment feature dimensionality.

LONGER-TERM DIRECTIONS

1. Probabilistic forecasting: predict a distribution over next segments rather than a point estimate; natural fit with diffusion or flow-based priors.
2. Physics-informed equations: add advection terms and Coriolis-parameterised ODE types to the library, bridging the gap between pure ML and NWP.
3. Graph extension: connect multiple stations via a graph neural network layer before the transformer, allowing spatial information to propagate.

9. Conclusion

We have proposed and implemented a novel two-stage approach to single-station weather forecasting: walk-forward ODE segmentation followed by transformer sequence prediction over analytical segment descriptors. On a 7-day test period at Knock Airport, Ireland, the approach achieves a **pressure MAE of 0.89 hPa at 12 h**, beating persistence by 34%, and **temperature MAE of 1.42°C at 24 h**, matching persistence with no atmospheric physics. The representation is interpretable, computationally lightweight, and extensible to additional channels and stations.

The primary limitation is the scope of evaluation: one week at one station is insufficient for a peer-reviewed claim of superiority over existing methods. The proposed research agenda outlines a path to a full paper with multi-station, multi-season validation and direct NWP comparison.

References

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